Measuring Coordination-Collapse Boundaries in Reward-Mixed Cooperative MARL

Anonymous submission

Abstract

Cooperative multi-agent reinforcement learning can settle into low-cooperation lazy-agent behavior when agents' local incentives dominate the team objective. We study not only whether reward centralization mitigates this failure, but where a trained population crosses into a pre-specified collapse region. We sweep the local-to-team reward mixture w, estimate $P(\text{collapse} \mid w)$, and report the midpoint w_{crit} with bootstrap confidence intervals; on our primary supply-chain testbed, $w_{\text{crit}} = 0.666$ $[0.635, 0.694]$. Using this boundary instrument, we evaluate a loss-averse reward transform that weights losses more heavily than equal-sized gains. On the primary testbed, we separate this effect from value-function curvature, which moves the boundary in the opposite direction. Across four environment–algorithm combinations, the loss-aversion direction persists, while longer training horizons move the boundary leftward. The thresholds are specific to the environment, learner, training budget, reward units, and collapse label, but the protocol provides a reusable instrument for locating coordination failure and measuring how reward transformations and learning conditions move its boundary.

Introduction

A team of learning agents can fail even when the task has a good cooperative solution: instead of learning complementary behavior, agents can settle into locally safe, low-cooperation behaviour. We call this *coordination collapse*; in the MARL literature its endpoint is the *lazy-agent* policy (Liu et al. 2023; Shi and Peng 2023), and the underlying private-versus-collective conflict is the one studied in sequential social dilemmas (Leibo et al. 2017). We study this failure through a *reward-centralization* dial: within each reward-mixing sweep, agent i is trained on the blended scalar $x_i(w) = (1-w)\, r_i^{\text{local}} + w\, r^{\text{global}}$, $w \in [0,1]$, so at $w{=}0$ the learner sees only its own local reward and at $w{=}1$ only the team reward. Turn w down and an agent's private performance drowns out its contribution; cooperating earns no learnable signal and the lazy policy wins. That decentralized reward shaping cannot match centralized training is broadly accepted (Akella 2025), and lazy-agent and credit-assignment work (Liu et al. 2023; Shi and Peng 2023; Sunehag et al. 2018; Rashid et al. 2018; Foerster et al. 2018) mitigates the pathology; what none of it supplies is the *location*. The closest phase-diagram studies place their critical points on *environmental* (Yamaguchi 2025) or opinion-dynamics (Capannoli et al. 2026) axes rather than reward structure, and none estimates a confidence-intervalled crossing in reward-centralization space for gradient-trained cooperative policies. Our central question is therefore twofold: does coordination collapse admit a *measurable* critical centralization threshold w_{crit}, and do prospect-theoretic reward transforms systematically shift it? The first is a measurement problem (the paper's primary contribution); the second is a question the instrument then makes answerable.

Pinning down this number matters because reward design is not neutral. w_{crit} is not universal: it is a measured property of an environment, learner, horizon, and collapse label that tells a designer where their own system crosses into collapse. We estimate it in four environment-algorithm combinations (a six-node inventory network under PPO and MAPPO, and Cooperative Navigation (MPE) and Level-Based Foraging (LBF) under MAPPO). Because the boundary is finite-horizon, we also re-measure it across horizons on the primary supply-chain testbed and the two transfer environments. The inventory network is a branching directed-flow graph, which later lets us ask *which* node's reward design sets the boundary. We use prospect theory only as a controlled reward-transform family, not a psychological claim: precisely, a pointwise loss-asymmetric transform of the immediate blended reward, omitting the probability-weighting channel of full prospect theory. After blending, the scalar is transformed with value-function curvature and loss aversion (a loss looming larger than an equal gain (Kahneman and Tversky 1979; Tversky and Kahneman 1992)), while the learner, architecture, optimizer, and evaluation label are unchanged. The experimental question is then clean: does changing the scalar training signal move the collapse boundary?

We build the instrument before using it. Three design choices make a reading trustworthy: the collapse label is a behavioral rule scored on evaluation and pre-registered on the primary supply-chain testbed (label, analysis code, and probes committed before execution); matched policies differ only in the scalar training signal, since the transform is ap-

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plied after blending, outside the optimizer; and every w_{crit} carries a bootstrap confidence interval. We use "boundary" in this operational sense: a finite-training, threshold-defined behavioral crossing in learning dynamics, not a claim of phase-like criticality.

The contributions are: (i) a confidence-intervalled protocol for locating collapse boundaries in reward-mixed cooperative MARL; (ii) a controlled reward-transform decomposition showing that value-function curvature and loss aversion can move the boundary in opposite directions, while the loss-aversion direction replicates across four environment–algorithm combinations and the finite-horizon slide recurs on the primary supply-chain testbed and both transfer environments; and (iii) a scoped heterogeneity result showing that, under parameter sharing on the supply chain, one loss-averse node can nearly set the team boundary, while a separate-actor control rules out a general topology-based ordering. Magnitudes and secondary structure are substrate-specific unless separately replicated. The contribution is not the logistic midpoint itself, but a comparative protocol for measuring how training choices and reward transformations move a system-level failure boundary, with uncertainty.

Related Work

Cooperative MARL, studied since independent Q-learning on shared tasks (Tan 1993; Buşoniu, Babuška, and De Schutter 2008), inherits the lazy-agent and relative-overgeneralization failures in which agents abandon cooperation for a degenerate policy (Liu et al. 2023; Shi and Peng 2023; Tampuu et al. 2017). The standard remedy is centralized credit assignment under CTDE, a centralized critic or factorized value function (Lowe et al. 2017; Yu et al. 2022; Sunehag et al. 2018; Rashid et al. 2018; Foerster et al. 2018; Son et al. 2019); related work ties the failure to decentralized exploration (Weng and Lee 2026) or shows decentralized shaping cannot match centralized baselines (Akella 2025). This establishes *that* centralization helps and *how* to implement it; none locates the threshold w_{crit} or gives it a confidence interval, the gap we close, using w as the dial rather than a fixed factorization. Combined individual-and-team reward design has direct precedent: DE-MADDPG separates a per-agent and a team critic for exactly such mixtures (Sheikh and Bölöni 2020), and packet-routing work has used RL routing as a testbed and later studied the lazy-versus-selfish trade-off with mixed and adaptive local/global signals (Boyan and Littman 1994; Mao, Gong, and Xiao 2020); but these set or adapt the mixture as a design choice rather than *locating* w_{crit} or giving it a confidence interval.

Treating collective learning as a dynamical system with critical points has precedent: Yamaguchi (2025) builds a phase diagram for independent Q-learning over *environmental* axes, and Capannoli et al. (2026) prove a metastable boundary in opinion dynamics under a fixed update rule. Mean-field and dynamical-systems analyses derive deterministic or distributional limits for multi-agent learning (Hu, Leung, and Leung 2019; Barfuss, Donges, and Kurths 2019; Leonte et al. 2025). We borrow the phase-diagram measurement style but differ in object: ours is a *reward-structure* threshold with confidence intervals, and we use the mean-field machinery only for a sign-level account (Supplement).

The heterogeneous-node result also has adjacent precedent in threshold and cascade models. Aggregate outcomes can depend on the distribution of actors' thresholds rather than their average preferences (Granovetter 1978), while small initial shocks can produce global cascades in susceptible networks (Watts 2002). We use this literature only as conceptual precedent, not as evidence of a cascade mechanism in our setting. Our contribution is to measure how assigning a loss-averse reward transform to a single node shifts a finite-training coordination-collapse boundary in gradient-trained cooperative MARL; moreover, a separate-actor control rules out a general topology-based ordering.

Human preference structure has entered collective learning only narrowly. Hughes et al. (2018) show inequity aversion improves cooperation within the cooperative-AI agenda (Dafoe et al. 2020; Hammond et al. 2025; Barfuss et al. 2020), and loss aversion is now a training objective at scale: KTO aligns language models with a Kahneman–Tversky value function carrying an explicit loss-aversion coefficient (Ethayarajh et al. 2024). Risk-sensitive RL optimizes tail outcomes (Qiu et al. 2021; Shen et al. 2023; Prashanth 2014; Tamar, Glassner, and Mannor 2015) (distinct from loss aversion's reference-point asymmetry), and prospect theory has entered single-agent RL with known calibration concerns (Prashanth et al. 2016; Rabin 2000; Tversky and Kahneman 1992). In multi-agent settings, Mengesha, Sun, and Roy (2025) find loss aversion *non-monotone* in an evolutionary quantal-response model, and Lalmohammed et al. (2025) bring CPT into cooperative MARL but fold it into the actor–critic update, varying value and probability-weighting components jointly with the optimizer. We differ at the level of *apparatus*: rather than proposing another behavioral model, we build a measured boundary and use it as a common instrument; the prospect-theoretic reward-transform study is its first application, not the instrument's whole contribution.

Preliminaries and Method

We study *reward-mixed* cooperative MARL, in which each learner's scalar reward is the blend $x_i(w) = (1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}$, $w \in [0,1]$; our primary supply-chain learner is parameter-shared PPO (a shared actor–critic with per-agent observations and decentralized execution; "IPPO" in the shared-parameter sense) and the transfer settings use a centralized critic. Here w is not an optimizer hyperparameter but a structural property of the reward design. We do not rescale r_i^{local} or r^{global} separately at each w; both stay in the environment's native units, so a given w means the same mixture in every condition rather than hiding a per-condition normalization. Unless stated otherwise, each experiment varies one factor at a time, holding the learner, environment, horizon, transform parameters, and label fixed.

The boundary estimate follows four steps: fix the environment, learner, horizon, and collapse label; train matched policies across a grid of w; label each run from pre-specified *evaluation* behavior, not training reward; and fit a monotone binary model $P(\mathrm{collapse} \mid w)$, reporting the midpoint w_{crit}

with bootstrap confidence intervals.

The reward transform is prospect-theoretic in value-function form: with zero as the gain/loss reference, $v(x) = x^\alpha$ for $x \geq 0$ and $-\lambda(-x)^\beta$ for $x < 0$. The exponents $\alpha, \beta \in (0, 1]$ control *curvature* (values below one make the transform sublinear, compressing larger magnitudes); the separate multiplier $\lambda > 1$ scales the entire loss branch, so a loss counts λ times an equal gain (*loss aversion*), and $\alpha{=}\beta{=}1, \lambda{=}1$ leaves $v(x) = x$. Because curvature is present in every loss-aversion condition, we run a curvature-only baseline so its effect is not silently folded into the λ shift. We omit the probability-weighting channel of full CPT, so only the scalar value transform varies between matched runs. Applied after blending,

$$r_i^{\mathrm{PT}} = v\big((1-w)\, r_i^{\mathrm{local}} + w\, r^{\mathrm{global}}\big)\,, \tag{1}$$

this keeps the optimizer and architecture bit-identical across matched conditions, the difference from Lalmohammed et al. (2025), who embed CPT *inside* the actor–critic update, varying transform, probability weighting, and optimizer together. A measured shift is thus attributed to the transform channel, while recognizing that reward scale, variance, and advantage shape are part of that channel.

A *collapse label* is a pre-specified binary evaluation rule marking runs whose behavior lies inside a low-cooperation region, scored on held-out evaluation episodes rather than training reward. In the supply chain, collapse is $\mathbf{1}[\bar{S} < 10 \wedge \bar{\pi} \leq -127.1]$, with $\bar{S}$ the mean base-stock and $\bar{\pi}$ the mean profit over deterministic evaluation; this corresponds to low/no ordering with poor profit. Collapsed runs pile up at the lazy-policy deterministic profit floor (≈ -128), so the profit clause is effectively a redundancy check on $\bar{S} < 10$ rather than a tuned cutoff. In MPE (Cooperative Navigation), collapse is landmark occupancy below $\theta = 0.144$, i.e. agents cluster rather than cover landmarks; in LBF (Level-Based Foraging), collapse is mean team return below $\theta = 0.626$ (the calibrated collapsed-versus-coordinated midpoint), i.e. agents fail to cooperatively load food. The supply-chain label and analysis code were frozen before execution; threshold robustness over estimable $\pm 20\%$ variants changes w_{crit} by at most 0.015 (Supplement).

For each condition we fit $P(\mathrm{collapse} \mid w) = \sigma(b_0 + b_1 w)$ and report $w_{\mathrm{crit}} = -b_0/b_1$, the w at which fitted collapse probability is 0.5, with percentile-bootstrap CIs over runs (Efron and Tibshirani 1993) (2,000 resamples). Where collapse is near-perfectly separated across adjacent w (steep high-λ cells) the ordinary slope diverges, so those cells use Firth's bias-reduced logistic regression (Firth 1993), whose w_{crit} agrees with the bootstrap to within 0.019. The location is not an artifact of the logistic link: a probit link and isotonic (nonparametric monotone) regression reproduce w_{crit} to within 0.02 at every λ on all three substrates, and the $\lambda{=}1{\to}7$ shift is unchanged under all three (Supplement). The transition width is the fitted 10–90 width $\Delta w = (\mathrm{logit}(0.9) - \mathrm{logit}(0.1))/|b_1|$. The decomposition compares, at identical environment and optimizer, a **linear baseline** ($\alpha{=}\beta{=}1, \lambda{=}1$), a **curvature baseline** ($\alpha{=}\beta{=}0.88, \lambda{=}1$), and **loss-aversion conditions** ($\alpha{=}\beta{=}0.88, \lambda > 1$). The primary substrate is `NetworkManagement-v1` (Hubbs et al. 2020), a six-node multi-echelon inventory network (Oroojlooyjadid et al. 2022; Gijsbrechts et al. 2022; Leluc et al. 2023), chosen because it exposes a directed-flow failure and supports the minority-topology analysis; transfer environments are MPE Cooperative Navigation (Lowe et al. 2017) and Level-Based Foraging (Christianos, Schäfer, and Albrecht 2020). Supply-chain agents run parameter-shared PPO (Schulman et al. 2017) via Stable-Baselines3 (Raffin et al. 2021); MPE, LBF, and the cross-algorithm control for supply chain use MAPPO (PPO with a centralized critic (Yu et al. 2022)) on the same stack, all wrapped with PettingZoo/SuperSuit (Terry et al. 2021; Terry, Black, and Hari 2020). Headline λ sweeps use 20 seeds per (w, λ) cell; the full run inventory is in the Supplement. Primary results are supply-chain/PPO; the supply-chain/MAPPO, MPE/MAPPO, and LBF/MAPPO settings are transfer tests of which claims generalize (Table 3).

The Boundary and Its Reward-Transform Shift

A well-defined boundary exists at fixed training budget. At 300k steps on the supply-chain PPO substrate, collapse has a clear, tightly-estimated boundary in reward-centralization space (Figure 1a): the fitted midpoint is $w_{\mathrm{crit}} = 0.666$ $[0.635, 0.694]$, and three independent estimators, logistic, probit, and isotonic (nonparametric monotone) regression, agree to within 0.02 (Method), a stronger check than any single link-function fit. The transition is monotone with a 10–90 width of $\Delta w = 0.199$ (about a fifth of the mixture axis); a Firth logistic-slope test also rejects the (much weaker) null of no relationship between w and collapse at all ($p = 3.2 \times 10^{-7}$): a resolved, non-flat crossing rather than phase-like sharpness. We use 300k as the primary horizon; horizon dependence is measured below.

"Collapse" is an operational threshold crossing, not a confirmed attractor. A fitted midpoint exists whenever the binary label varies with w, so we tested the underlying continuous behavior for a distinct dynamical regime: bootstrap 95% CIs on Sarle's bimodality coefficient (Sarle 1983) and Hartigan's dip test (Hartigan and Hartigan 1985) against a unimodal null, across both substrates and all w (Supplement). On the supply chain Sarle's coefficient rises near the crossing (0.59 at $w{=}0.65$ and 0.57 at $w{=}0.75$, both above the 0.555 rule-of-thumb) but its CI straddles the threshold everywhere, and the dip test never rejects unimodality ($p \geq 0.30$). LBF's raw team return shows nominally significant dip statistics at high w ($p < 0.05$), but this traces to a ceiling effect (runs saturating at the metric's upper bound), not a second basin: excluding ceiling-clipped runs removes the significance ($p \geq 0.11$). Neither substrate clears a formal multimodality test at our sample sizes, so we drop the attractor language and use *threshold-defined behavioral crossing* throughout.

The boundary is finite-horizon. The boundary slides left with the training horizon: across four matched risk-neutral horizons $w_{\mathrm{crit}} = 0.822, 0.666, 0.622, 0.573$ at $150/300/450/900$k steps, log-linear in T ($w_{\mathrm{crit}} = 2.44 - 0.138 \ln T$, RMSE 0.027; all four CIs of width ≤ 0.073; Figure 1b). Longer training rescues some runs that collapse at

shorter horizons; at any fixed budget w_{crit} is well-defined, so horizon dependence does not make it arbitrary. The log-linear fit is local to the tested range (150k–900k, a single order of magnitude); we use it to summarize the measured slide, not to extrapolate an asymptotic boundary. Mechanistic escape interpretations are treated only as supplementary, non-load-bearing explanations.

The horizon slide is not a supply-chain artifact. Under centralized-critic MAPPO the same slide recurs in both transfer environments: on MPE w_{crit} falls $0.82 \to 0.68$ over $2\text{M} \to 8\text{M}$ steps, and on LBF $0.33 \to 0.20$ over $1\text{M} \to 4\text{M}$, both with non-overlapping endpoints. A basin control rules out the confound that runs collapse because they start closer to the lazy basin: within each w, initial-to-converged base-stock correlations are only small to moderate (-0.11 to -0.38) and collapsed versus surviving runs are not separated by their start (Supplement).

Loss aversion raises the boundary; curvature lowers it. Two baselines anchor the comparison, all with identical environment, learner, label, and 300k horizon. The linear null ($\alpha{=}\beta{=}1, \lambda{=}1$) was measured twice as a reproducibility check (0.666 $[0.635, 0.694]$ in the main structural sweep and 0.685 $[0.657, 0.710]$ in the independent decomposition run, overlapping CIs consistent with one underlying value; we anchor the decomposition on 0.685, its own run); the second is the curvature baseline ($\alpha{=}\beta{=}0.88, \lambda{=}1$), fixed at the conventional prospect-theory curvature estimate (Tversky and Kahneman 1992), not tuned; the sweep then varies only $\lambda \in \{1, \ldots, 7\}$. The boundary rises monotonically up to noise, saturating near 0.85 with clear headroom to the $w{=}1$ ceiling (so the plateau is transform saturation, not a ceiling artifact; endpoints non-overlapping by 0.221; Figure 2, Table 1). Curvature alone acts in the *opposite* direction, lowering w_{crit} to 0.585 (-0.100 $[-0.172, -0.033]$ vs. linear), which is why we decompose: as a single "prospect-theoretic" treatment the two effects partially cancel, since here curvature eases coordination while loss aversion hinders it.

Two effect sizes answer two questions. The effect size depends on the baseline (Table 1). The *deployment shift* against the linear baseline is the total change a designer sees adopting the full transform; the *isolated loss-aversion shift* against the curvature baseline is the cleaner estimate of the λ effect, since curvature is present in every sweep condition, and the two differ by the curvature offset (-0.100). Throughout, "the loss-aversion effect" means the increment over the curvature baseline. A one-parameter saturating curve (Supplement, a descriptive fit) summarizes the full series to RMSE 0.014.

The shift survives reference-point, scale, and reward-unit controls. Because λ also changes reward scale and the gain–loss split, we ran three matched controls on the supply chain (identical seeds, w-grid, optimizer; full design in the Supplement). (i) *Reference point*: re-centering the gain–loss reference from 0 to the empirical mean (-2.99), median (-6.77), or -0.5σ (-5.14) preserves the $\lambda{=}1{\to}7$ rise in every case ($+0.228$, $+0.183$, $+0.193$; all bootstrap CIs exclude zero, vs. $+0.269$ at reference 0). (ii) *Reward-scale match*: normalizing the transformed reward to a common RMS

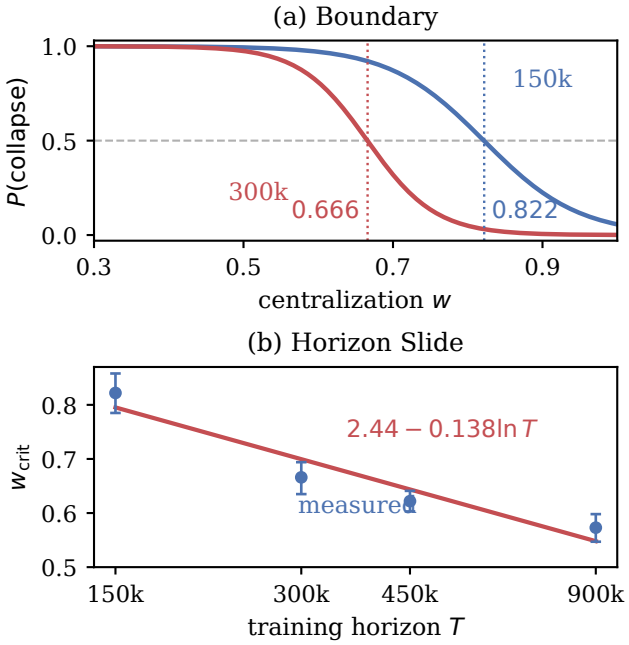


Figure 1: Boundary location and horizon slide. (a) At each fixed training budget, collapse probability changes steeply with reward centralization w; curves show fitted $P(\text{collapse} \mid w)$ at two representative horizons (150k and 300k steps). (b) All four measured horizons: increasing the training horizon shifts w_{crit} left, log-linearly in T (RMSE 0.027; 20 seeds per w-cell at each horizon), so more training rescues some runs that collapse at shorter horizons.

across λ (removing the $\approx 4\times$ magnitude growth from $\lambda{=}1$ to 7) leaves a significant residual shift ($+0.145$ $[0.107, 0.185]$), so the effect is not reward amplification alone. (iii) *Reward-unit sensitivity*: doubling the local reward before blending moves the baseline boundary to 0.779 and doubling the team reward to 0.476, yet the λ direction persists in both arms ($+0.130$ $[0.105, 0.153]$ and $+0.329$ $[0.263, 0.430]$). Because absolute w_{crit} depends on reward units, we also compute the normalized incentive ratio $\varrho = (1 - w_{\text{crit}})c_\ell / (w_{\text{crit}}\, c_g)$ from the reward-unit sweep (values and CIs in the Supplement). ϱ is not itself fully invariant across arms, but it is tighter than raw w_{crit}: at $\lambda{=}1$ the reward-unit spread shrinks from 0.30 in w_{crit} to 0.16 in ϱ (roughly halved; the same holds, more weakly, at $\lambda{=}3, 7$). Its *direction* is preserved: ϱ falls monotonically with λ in every arm, so the loss-aversion shift is not a reward-unit artifact even though ϱ's absolute level is partly one. Absolute-w_{crit} CIs are percentile bootstraps over whole runs; every difference is a seed-paired bootstrap over the shared seed identifier spanning all w-cells (2,000 resamples). The same integer seed fixes PPO's initialization, environment RNG, and evaluation episodes, so paired differences compare matched training/evaluation randomness; a broader unpaired-bootstrap check gives near-identical estimates and only marginally wider CIs (Supplement).

The curvature result is scoped more narrowly: on the supply-chain PPO substrate, curvature and loss aversion

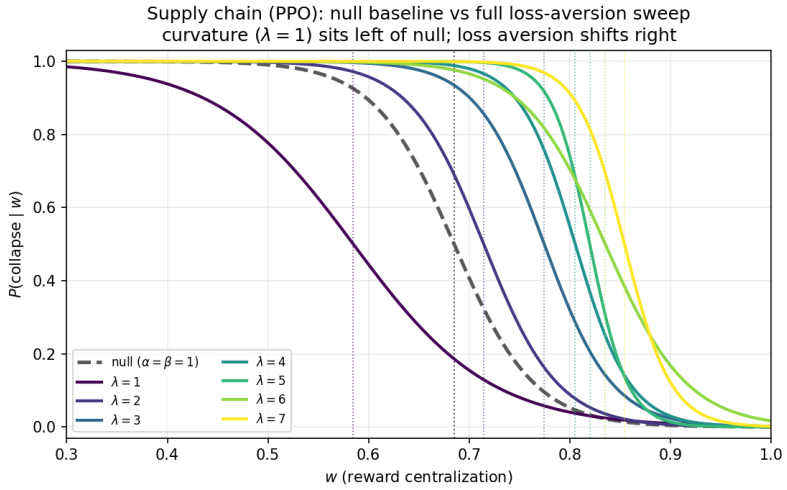


Figure 2: Loss aversion raises the collapse boundary (supply chain, PPO). Curves are fitted $P(\text{collapse} \mid w)$; dotted lines mark w_{crit}. Curvature alone ($\alpha{=}\beta{=}0.88$, $\lambda{=}1$) shifts the boundary *left* of the linear baseline; increasing λ then shifts it *right*, saturating near $w_{\text{crit}} \approx 0.85$.

Condition	α, β	λ	w_{crit}	vs. null	vs. curv.
Linear null (rep.)	1, 1	1	0.685	–	–
Curvature	0.88, 0.88	1	0.585	$-0.100^{\checkmark}$	–
Loss av.	0.88, 0.88	3	0.774	$+0.089^{\checkmark}$	$+0.189^{\checkmark}$
Loss av.	0.88, 0.88	7	0.854	$+0.169^{\checkmark}$	$+0.269^{\checkmark}$

Table 1: Decomposing reward-transform shifts (supply chain, PPO, 300k). "vs. null" is the deployment shift against the reproduced linear baseline; "vs. curv." is the isolated loss-aversion shift against the curvature baseline. ✓ = difference CI excludes zero.

move w_{crit} in opposite directions (Table 1); the transfer experiments show the loss-aversion direction generalizes while the curvature sign does not. A speculative, non-load-bearing account of *why* the sign goes this way, an escape signal-to-noise argument, a sign proposition resting on an unmeasured premise, an exploration-noise dissociation, and a single-substrate summary curve, is given in full in the Supplement; none of it is required for, or changes, the measurements above.

The Instrument Travels; So Does the Direction

Beyond the supply chain, we estimate a CI'd w_{crit} in three further settings: the supply chain under MAPPO (changing only the learner), and MPE and LBF under MAPPO (changing the environment). The design is not fully crossed (parameter-shared PPO floors out on the shared-reward transfer tasks), so the supply-chain/MAPPO cell is what separates the learner change from the environment change. In each we ask whether the loss-aversion direction travels.

A transfer boundary is meaningful only if the learner can coordinate at high centralization. Independent PPO fails this on the shared-reward transfer tasks: it collapses at *every* w on both MPE and LBF (a coordination floor; Supplement), right-censoring the boundary rather than removing it, whereas centralized-critic MAPPO (Yu et al. 2022) recovers coordination and exposes an interior boundary. MPE has no native local–global conflict, so we construct one with a herding wrapper (local reward for staying near others) and fix its scale, before the λ-sweep, so the linear no-transform boundary is interior ($w_{\text{crit}} = 0.74$). MPE therefore tests whether the boundary instrument and loss-aversion direction travel after changing the environment and learner; the absolute thresholds remain conditional on this fixed wrapper and scale. Collapse is landmark occupancy below $\theta = 0.144$.

The loss-aversion direction transfers in every setting (Figure 3, Table 3), reported at each setting's largest interior λ ($\lambda{=}7$ for supply and MPE, $\lambda{=}4$ for LBF, which saturates against $w{=}1$ beyond that). On the *supply chain under MAPPO* (only the learner changes) w_{crit} rises $0.796 \rightarrow 0.889$, converging with PPO by $\lambda \geq 4$; on *MPE* $0.78 \rightarrow 0.89$ ($+0.11$); on *LBF* $0.26 \rightarrow 0.83$ by $\lambda{=}4$. Secondary structure does not transfer: on MPE the curvature shift spans zero ($+0.042$), and on LBF curvature moves the boundary *right* while the summary form misfits (RMSE 0.15); so the portable claim is the CI'd boundary and the loss-aversion direction, not the levels, curvature sign, or summary form.

Scope condition: action representation (single granularity tested). On the base-stock control the boundary depends on action granularity. Value-based learners (QMIX, IQL) on a discretized $K{=}11$ shell give a flat collapse rate, but PPO on the *same* shell also goes flat (Table 2), so it is coarse discretization, not the value-based paradigm, that erases the boundary; MPE, natively discrete yet retaining a boundary, shows this is coarseness rather than discreteness per se. We tested only $K{=}11$, so we cannot yet distinguish the boundary reappearing gradually as K grows from it requiring near-continuous control specifically, which would scope out value-based learners more fundamentally than "coarse dis-

Learner	Action space	Boundary?
PPO / MAPPO	continuous	yes
QMIX (centralized value)	discrete $K{=}11$	no
IQL (independent value)	discrete $K{=}11$	no
PPO (control)	discrete $K{=}11$	**no**

Table 2: Action-representation ablation. The last row is the key control: it changes only the action space relative to continuous-control PPO, isolating coarse discretization as what erases the boundary.

Setting	$w_{\mathrm{crit}}(\lambda{=}1)$	w_{crit}(last)	Δ
Supply PPO	0.585 [.545, .626]	0.854 [.837, .871]	+0.269
Supply MAPPO	0.796 [.742, .860]	0.889 [.856, .917]	+0.093
MPE MAPPO	0.78 [.73, .83]	0.89 [.84, .93]	+0.11
LBF MAPPO	0.26 [.20, .32]	0.83 [.78, .89]	+0.57

Table 3: Generalization of the loss-aversion *direction*. Within-sweep endpoints ($\alpha{=}\beta{=}0.88$) with bootstrap 95% CIs, $\lambda{=}1 \rightarrow$last interior (LBF: $\lambda{=}4$); Δ is the endpoint difference.

cretization" suggests. A K-sweep ($K \in \{11, 25, 51, 101\}$) would resolve this; untested here, so the loss-aversion *direction* under any value-based learner remains an open scope question, not a tested negative.

A Single Downstream Node Can Nearly Set the Boundary

This experiment is necessarily supply-chain-only: it requires a directed resource-flow topology (a DAG) that MPE and LBF lack. The main sweep is homogeneous; here we make one node loss-averse while the others stay risk-neutral, holding everything else identical to the main sweep, and ask whether one biased node can move the boundary and whether its position sets the shift. Position could matter because the network is a branching DAG: demand enters at the retail edge, two downstream nodes feed it directly, and upstream suppliers feed one or both paths, so a node should matter more when its output reaches demand through a short, concentrated path. Because MPE and LBF lack this topology, the heterogeneous-node effect is tested here only on the supply chain.

A single node, dose for dose. Sweeping the most downstream node's $\lambda \in \{1, \ldots, 7\}$ (others risk-neutral) traces a strictly monotone dose-response: w_{crit} rises $0.585 \rightarrow 0.863\,[0.846, 0.880]$, matching the *fully* loss-averse (homogeneous) value ($0.854\,[0.837, 0.871]$) within noise (CIs overlap) by $\lambda{=}7$ with the same saturating shape as the homogeneous sweep. Operationally, the team's collapse boundary is not an average over agents: a single node's reward design can nearly set it on its own, so an audit of reward transforms should not treat the boundary as an average over agents. Which node matters most is position-dependent under a shared actor but does not survive as a general rule once the shared gradient is removed (below).

Position: proximity to demand, not chain index, under a shared actor. Placing one $\lambda{=}7$ node at each position, the effect is strongly position-dependent, and under the shared-parameter actor the observed ordering is consistent with how directly a node's output reaches demand, not a chain index. The two retail-feeding nodes dominate (0.863, 0.800); the sole supplier to the dominant downstream node is comparable within noise (0.797); the split-path suppliers are weaker (0.726, 0.673, also overlapping each other). Because a single shared actor can let a biased node's gradient influence others' behavior directly, and because nodes at different positions encounter different observations, visitation, and advantage distributions even under a common observation format, this ordering is not by itself evidence that the mechanism is topological: we designed a second, independently-parameterized sweep specifically to test this before drawing that conclusion.

The ordering does not survive without a shared gradient. We repeat the identical position sweep with five fully independent PPO actors, one per trainable node, each updated only on its own transitions and sharing no parameters or gradient with the others (same env, w-grid, seeds, α, β, λ, label, and estimator; each actor trains for 60k real environment ticks, matching the shared actor's per-agent exposure). This is not a reshuffling: the previously dominant node (0.863, rank 1) becomes the weakest of the five ($0.498\,[0.363, 0.581]$, rank 5), while the previously weakest node (rank 5) rises to rank 2 (Table 4). The coarse retail-adjacent-vs-split-path grouping also fails to replicate (mean 0.832 vs. 0.700 under the shared actor; 0.606 vs. 0.553, no longer cleanly separated, under separate actors). We conclude the position-specific ordering under the shared actor is substantially a shared-gradient artifact rather than evidence for a directed product-flow mechanism, and we do not draw a topological reading from it. Where Mengesha, Sun, and Roy (2025) find loss aversion's effect on cooperation is *non-monotone* in an evolutionary model, the sign of the effect here (loss aversion raises w_{crit}) is monotone and, unlike the position ordering, is not an artifact of parameter sharing (Table 1).

Discussion

To our knowledge no prior work separates the transform's curvature and loss-aversion components in cooperative MARL (Lalmohammed et al. 2025; Ghaemi et al. 2024). On the supply-chain PPO substrate the two move the boundary in *opposite* directions and would partially cancel if reported jointly, so a single "prospect-theoretic" condition under-reports the loss-aversion effect by about half there; the curvature sign itself is supply-chain-specific and not measured to transfer: it spans zero on MPE and runs the *same* direction as loss aversion on LBF.

The boundary is a design quantity: a team can budget for the loss-aversion margin rather than discover collapse in production: if w is capped at, say, 0.8, the usable budget is

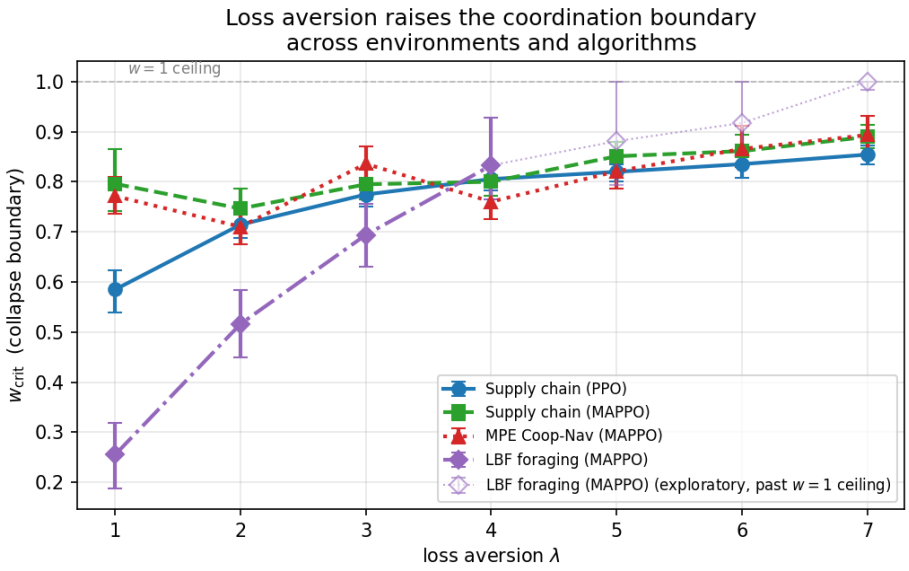


Figure 3: w_{crit} versus λ across four settings: supply chain under PPO and MAPPO, MPE under MAPPO, LBF under MAPPO (bootstrap 95% CIs). w_{crit} rises with λ in every setting; the *direction* generalizes, while levels (especially at λ=1), curvature sign, and exact form are substrate-specific. LBF $\lambda \geq 5$ points extrapolate past the w=1 ceiling (outside the measured grid); they are drawn as hollow markers on a dotted line, visually distinct from the solid in-grid points, and are exploratory only.

Position	Shared	Rank	Separate	Rank
node_2	0.863 [.846, .880]	1	0.498 [.363, .581]	5
node_3	0.800 [.780, .820]	2	0.714 [.683, .745]	1
node_5	0.797 [.740, .857]	3	0.538 [.474, .591]	4
node_6	0.726 [.682, .769]	4	0.547 [.495, .594]	3
node_4	0.673 [.622, .720]	5	0.559 [.502, .608]	2

Table 4: Position sweep, shared vs. separate actors (λ=7, n=20/position/actor condition). Rank 1 = strongest effect on w_{crit}. Shared and separate columns report w_{crit} with bootstrap 95% CIs.

where w_{crit} crosses that cap, about $\lambda \approx 4$ here. The minority result shows a single node's reward design can nearly set the whole team's boundary, so an audit of reward transforms should not treat the boundary as an average over agents; identifying *which* node matters most is system-specific (the deconfounded sweep above rules out a general position rule) and whose dual-use we address in the ethics statement.

Scope and Future Work. Three limits point to next steps. The quantitative core is single-substrate (magnitudes belong to one simulator, the summary form does not transfer), and the minority-node sensitivity is not cleanly a directed-flow effect (the deconfounded sweep rules out a general position rule on this substrate), so the open question is narrower than a second directed-flow domain: whether minority-agent boundary dominance under parameter sharing transfers across networks and learning architectures, not whether a positional cascade does. The mechanism fixes sign but not magnitude: localizing w_{crit} from dynamics resisted two pre-committed probes, so a localizing observable is the clearest open problem. And ϱ only partially compensates for reward-unit dependence; finding a fully unit-invariant coordinate, or showing none exists for this class of transforms, is as open a problem as localizing w_{crit} itself.

Conclusion

Reward centralization is usually treated as a direction: more team signal helps. We have shown it also has a measurable location, a critical mixture w_{crit}, that moves rightward under loss-averse transforms and leftward with longer training, and that a single node's reward design can nearly set it on its own under parameter sharing, a sensitivity that a deconfounded, independently-parameterized sweep shows is not a general topological rule. The direction and horizon slide generalize; the thresholds are system-specific, but the protocol is not.

Ethical Statement

This work is entirely simulated, with no human subjects or personal data. Reported thresholds are substrate-specific and must be re-measured before use; the minority-node result is dual-use (hardening vs. targeted disruption), reported explicitly for auditability.

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